ERES INSTITUTE FOR NEW AGE CYBERNETICS

White Paper

Smart-Resonant Offset Contracts (SROCs)

subject to

NBERS · UBIMIA · Gracechain

SOMT @ ERES Institute for New Age Cybernetics

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Game Theory + 1000-Year Future Map

Anchoring Document: How = International Land Ownership Reference: Personal / Public-Private

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Don't hurt yourself. Don't hurt others. Build for generations to come.

$$C = R \times P / M$$

Abstract

This white paper specifies the architecture, game-theoretic foundations, and operational mechanics of Smart-Resonant Offset Contracts (SROCs) within the ERES New Age Cybernetics framework. SROCs are cybernetically governed agreements that bind land stewardship, resource flows, and economic exchange to the 1000-Year Future Map's viability constraints.

SROCs operate subject to three interlocking subsystems: NBERS (the bio-ecologic rating system that scores every tile and actor for ecological integrity), UBIMIA (the merit-based universal income mechanism that distributes economic returns proportional to stewardship performance), and Gracechain (the blockchain ledger that records, validates, and enforces contract execution with cryptographic finality).

The paper demonstrates how SROCs, when anchored to the How = International Land Ownership Reference (Personal/Public-Private) and tested through game-theoretic equilibrium analysis, produce a self-correcting governance mechanism where cooperation dominates defection across civilizational time horizons, and where the SOMT (Societal Operational Management Tapestry) provides the planetary coordination layer.

1. Introduction

The global land governance crisis is not a failure of information but a failure of method. Existing land-ownership frameworks are fragmented across jurisdictions, temporally myopic, and structurally incapable of integrating ecological, social, and economic signals into a unified governance loop. The result: watershed degradation, displacement cascades, speculative extraction, and civilizational brittleness at every scale.

The ERES Institute's How = International Land Ownership Reference (ERES-HOW-ILO-2026-001) established the foundational triad—Personal, Public, Private—as the stewardship classes for all land governance under the 1000-Year Future Map. This white paper extends that foundation by specifying the contractual mechanism through which the triad is enforced, incentivized, and dynamically maintained: the Smart-Resonant Offset Contract (SROC).

An SROC is not a conventional smart contract. It is a cybernetically governed agreement that resonates—meaning its terms, execution, and outcomes are continuously validated against the ARI/RHC loop to confirm alignment with the 1000-year viability map. When an SROC detects deviation, it does not simply penalize; it offsets—triggering remediation pathways (NPR) that restore the system to green-band operation.

1.1 Scope

- Define the SROC architecture and lifecycle within PlayNAC
- Establish NBERS, UBIMIA, and Gracechain as the three subject-matter constraints governing all SROCs
- Map SROCs to the How triad (Personal/Public-Private) and the three 1000-Year Future Map scenes
- Apply game-theoretic analysis to demonstrate equilibrium stability across multi-generational horizons
- Specify the SOMT integration pathway for planetary-scale coordination

2. Foundational Architecture

2.1 What Is an SROC?

A Smart-Resonant Offset Contract (SROC) is a formalized agreement between two or more PlayNAC actors (persons, agencies, tiles, organizations, or governance bodies) that encodes:

- Terms: what resources, rights, or duties are exchanged
- Conditions: what EarnedPath and NBERS scores must be maintained for the contract to remain active
- Resonance Binding: the ARI/RHC validation loop that continuously tests contract execution against the 1000-year viability map
- Offset Mechanism: the NPR-aligned remediation pathway that activates when any party or tile deviates from green-band metrics

Smart: Computationally executable on Gracechain with deterministic state transitions.

Resonant: Validated through ARI/RHC loops—not merely legally compliant but harmonically aligned with civilizational constraints.

Offset: Deviation triggers remediation, not punishment. The contract self-corrects toward the green band through NPR protocols.

Contract: A binding agreement with cryptographic finality on Gracechain, enforceable across jurisdictions through the SOMT coordination layer.

2.2 The Three Subject-Matter Constraints

Every SROC compiles against three subsystems. If any constraint is violated, the SROC enters offset mode.

2.2.1 NBERS (Bio-Ecologic Rating System)

NBERS scores every tile and actor for ecological integrity across multiple dimensions: soil health, water quality, biodiversity indices, carbon flux, and atmospheric resonance. An SROC's ecological clause must specify the NBERS bands that all participating tiles must maintain. If a tile's NBERS score degrades below the contract's specified floor, the offset mechanism activates—triggering ecological remediation funded by the contract's offset reserve.

2.2.2 UBIMIA (Universal Basic Income through Merit, Innovation, and Achievement)

UBIMIA ensures that the economic returns generated by SROCs are distributed proportional to stewardship merit, not extractive capacity. Every SROC must include a UBIMIA distribution clause specifying how value flows to actors based on their EarnedPath scores. This prevents SROCs from becoming vehicles for concentration—the merit-based distribution function guarantees that productive stewardship is rewarded and extractive behavior is economically disadvantaged.

2.2.3 Gracechain (GCF — Gracechain Consensus Framework)

Gracechain is the distributed ledger that records SROC state, validates transitions, and enforces execution with cryptographic finality. Gracechain operates on a consensus model aligned with Meritcoin economics—transaction validation is weighted by stewardship merit (EarnedPath scores), not computational power or token holdings. Every SROC state transition is recorded as an immutable Gracechain entry, creating a permanent audit trail for the 1000-year viability map.

3. SROC Mapping to the How Triad

The How = International Land Ownership Reference defines three stewardship classes: Personal, Public, and Private. Each class generates a distinct SROC type with specific constraint profiles.

SROC Type	How Class	Primary Scene	NBERS Role	UBIMIA Role
SROC-P	Personal	BORDER_MM (micro)	Household ecological floor	Dignity income baseline
SROC-U	Public	WIS_PT (permanent)	Watershed/commons integrity	Public stewardship dividend
SROC-V	Private	PROP_LL (conditional)	Productive-use green band	Merit-proportional returns

3.1 SROC-P: Personal Stewardship Contracts

SROC-P contracts govern the relationship between a person/household and their tile. They encode micro-border flow rules (who and what crosses the threshold), ecological maintenance duties, and the dignity guarantees from the WIS_PT permanent layer. The UBIMIA clause ensures a baseline income floor tied to the person's EarnedPath score—no one falls below dignity while maintaining stewardship.

3.2 SROC-U: Public Commons Contracts

SROC-U contracts govern commons: watersheds, migration corridors, ecological reserves, and shared infrastructure. These are the most heavily constrained SROCs—their terms are anchored to the PlayNAC KERNEL's permanent layer and cannot be overridden by any single actor. NBERS scores for public tiles have the highest floor thresholds. UBIMIA distributes a public stewardship dividend to all actors who contribute to commons maintenance, creating a direct economic incentive for collective care.

3.3 SROC-V: Private Venture Contracts

SROC-V contracts govern productive private use of tiles. They encode the "right plus duty" principle from PROP_LL: extraction privileges are conditional on maintaining green-band metrics across ecological, social, and justice dimensions. UBIMIA returns are merit-proportional—the better the stewardship score, the higher the economic return. Tiles falling below ARI thresholds trigger NPR remediation funded by the contract's offset reserve, not punitive seizure.

4. Game-Theoretic Foundations

SROCs are designed so that cooperation is the dominant strategy across all time horizons. This section formalizes the game-theoretic properties that make defection (extraction, enclosure, commons destruction) structurally irrational under the SROC regime.

4.1 The Stewardship Game

Model each tile-steward relationship as an iterated game with the following properties:

- Players: tile stewards (Personal, Public agents, Private enterprises)
- Strategies: Cooperate (maintain green-band metrics, honor SROC terms) or Defect (extract beyond thresholds, degrade commons, violate contract)
- Payoffs: determined by EarnedPath scores, UBIMIA distribution, and Gracechain-recorded reputation
- Iteration: infinite-horizon (the 1000-year map ensures no known terminal round)

4.1.1 Why Cooperation Dominates

In classical game theory, cooperation in finite games is unstable because players can defect in the final round without consequence (backward induction). The SROC regime eliminates this instability through three mechanisms:

1. Infinite Horizon: The 1000-year map has no terminal round. No actor can calculate a “last move” advantage because the game continues beyond any individual lifespan, and EarnedPath scores persist across generational transfers.
2. Reputation Persistence: Gracechain records are immutable. Defection permanently degrades an actor’s stewardship reputation, reducing UBIMIA returns and future SROC eligibility across all tiles and jurisdictions.
3. Offset over Punishment: NPR remediation means defection triggers restoration costs, not destruction. The defector’s tiles enter remediation, reducing their productive capacity without annihilating their position—making “defect and rebuild” economically irrational compared to sustained cooperation.
1. Commons Shield: Public tiles (SROC-U) are permanently non-alienable. No accumulation of private power can acquire the commons, eliminating the “monopoly endgame” that destabilizes most property regimes.

4.2 Nash Equilibrium Analysis

Under the SROC regime, the unique Nash Equilibrium is mutual cooperation. The proof follows from three properties:

- No unilateral deviation improves payoff: defecting reduces EarnedPath scores, UBIMIA returns, and Gracechain reputation simultaneously
- No coalition of defectors can sustain advantage: NBERS monitoring detects ecological degradation before it cascades, and NPR offsets restore tiles before permanent damage
- The discount factor approaches 1: because the 1000-year horizon makes future payoffs nearly as valuable as present payoffs, the shadow of the future is maximally long

4.3 Mechanism Design: Incentive Compatibility

SROCs satisfy the incentive-compatibility constraint: truth-telling and cooperative behavior maximize each actor's expected utility. This is achieved through:

- Revelation Principle: NBERS sensors and ARI/RHC loops automatically reveal tile state—actors cannot misrepresent ecological or social conditions
- Individual Rationality: UBIMIA guarantees a dignity floor, ensuring participation is always preferred to exit
- Budget Balance: Gracechain's Meritcoin economics ensure that offset costs are funded from contract reserves, not external subsidies—the system is self-financing

5. SOMT Integration: Planetary Coordination Layer

The Societal Operational Management Tapestry (SOMT) is the planetary-scale coordination architecture that weaves SROCs into a coherent governance fabric. Without SOMT, SROCs would operate as isolated contracts; with SOMT, they become threads in a civilizational tapestry that spans from household tiles to interplanetary corridors.

5.1 SOMT Functions for SROCs

- Cross-Jurisdictional Compilation: SOMT ensures that SROCs in one LOGOS-region do not contradict constraints in adjacent regions, watersheds, or migration corridors
- Tier Integration: SOMT maps SROCs across the GSSG network tier architecture (T1–T8), ensuring that local contracts compile against planetary and interstellar constraints
- Temporal Coordination: SOMT maintains the 1000-year viability timeline, triggering re-validation cycles when accumulated SROC activity risks long-horizon degradation
- Seed Capsule Architecture: For interstellar-scale SROCs, SOMT's seed capsule metadata ensures that contract terms survive transit and remain valid upon arrival at destination systems

5.2 SOMT × Game Theory: Multi-Level Equilibrium

SOMT enables multi-level game-theoretic analysis. At each tier, the Nash Equilibrium of cooperation holds, and SOMT's compilation rules ensure that equilibria at lower tiers do not destabilize equilibria at higher tiers. This produces a nested equilibrium structure:

- Tile Level (T1): SROC-P/V contracts maintain household and enterprise cooperation
- Corridor Level (T3–T4): SROC-U contracts maintain watershed and migration-corridor cooperation
- Planetary Level (T5–T6): SOMT compilation ensures no regional SROC portfolio degrades planetary NBERS scores
- Interstellar Level (T7–T8): SOMT seed capsules carry SROC templates for colonial governance that inherit Earth-validated equilibria

6. SROC Lifecycle

Every SROC passes through a defined lifecycle managed by PlayNAC scenes and recorded on Gracechain.

6.1 Phase 1: Proposal

An actor proposes an SROC by submitting terms to the PlayNAC KERNEL. The proposal must specify: participating actors, affected tiles, stewardship class (Personal/Public/Private), NBERS floor thresholds, UBIMIA distribution formula, and offset reserve allocation. The KERNEL compiles the proposal against the 1000-year viability map.

6.2 Phase 2: Resonance Validation

The ARI/RHC loop evaluates the proposed SROC for harmonic alignment with existing contracts, tile states, and long-horizon constraints. Proposals that pass receive a Resonance Clearance Score (RCS). Proposals that fail receive specific deviation reports identifying which constraints are violated.

6.3 Phase 3: Execution

Cleared SROCs are deployed to Gracechain with cryptographic finality. Meritcoin staking by participating actors activates the contract. State transitions are recorded in real time as NBERS sensors, EarnedPath scoring, and ARI/RHC loops continuously monitor performance.

6.4 Phase 4: Offset (if triggered)

When any parameter deviates below the contract's specified thresholds, the SROC enters offset mode. NPR remediation protocols activate: the offset reserve funds restoration work, affected actors receive adjusted EarnedPath scores, and the contract enters a probationary cycle with enhanced RHC monitoring. Offset is not punishment—it is healing.

6.5 Phase 5: Renewal or Graceful Termination

SROCs have defined review cycles (aligned with RHC temporal harmonics). At each review, the contract is re-validated against current tile states and 1000-year projections. Contracts that remain in green band are renewed with updated terms. Contracts that cannot be restored are gracefully terminated, with tile stewardship reassigned through the How triad's compilation hierarchy.

7. 1000-Year Future Map Integration

The 1000-Year Future Map is the top-layer constraint against which all SROCs must compile. This section maps each Future Map scene to its SROC governance function.

Scene	SROC Types	Constraint Function	Game-Theoretic Role
BORDER_MM	SROC-P, SROC-U	Micro/macro border flow rules; jurisdictional inheritance	Defines player scope and strategy space at each scale
PROP_LL	SROC-V, SROC-P	Geospatial ledger; conditional stewardship scoring	Payoff function — EP scores determine returns
WIS_PT	SROC-U, all types	Permanent guardrails; temporary emergency scripts with decay	Commitment device — non-derogable rules prevent defection endgames

7.1 The Viability Compilation Rule

Every SROC portfolio across the SOMT tapestry must satisfy the following compilation hierarchy (inherited from the How specification):

1. Public permanent constraints (WIS_PT) are tested first
2. Private conditional duties (PROP_LL) are validated second
1. Personal micro-rules (BORDER_MM) are checked third

If any SROC in the portfolio violates a higher-tier constraint, it fails ARI/RHC validation and enters offset mode. The 1000-year viability score of the entire portfolio is recalculated with each SROC state transition, ensuring continuous alignment.

8. Economic Model: Meritcoin × UBIMIA × SROC

The economic engine of the SROC regime operates through the integration of Meritcoin (the unit of account), UBIMIA (the distribution mechanism), and the SROC offset reserve (the self-financing mechanism).

8.1 Value Flow

- Meritcoin Generation: Meritcoin is generated through validated stewardship activity recorded on Gracechain. Mining is merit-based—actors with higher EarnedPath scores generate more Meritcoin per unit of stewardship activity
- UBIMIA Distribution: Meritcoin flows through UBIMIA channels proportional to stewardship merit. The dignity floor ensures a baseline for all participants; the merit multiplier rewards excellence
- Offset Reserve: Every SROC allocates a percentage of value flow to its offset reserve. This reserve funds NPR remediation when tiles deviate from green-band metrics
- Gracechain Transaction Fees: Minimal fees fund network maintenance and SOMT coordination infrastructure

8.2 Anti-Extraction Guarantee

The SROC economic model structurally prevents extraction cascades. Because UBIMIA distribution is merit-proportional and NBERS scores gate contract eligibility, no actor can accumulate disproportionate economic power without proportionally exceeding stewardship expectations. Extraction—taking value without maintaining stewardship—automatically degrades the extractor's EarnedPath score, reduces their UBIMIA returns, and eventually triggers SROC offset mode on their tiles.

9. Security Architecture

SROCs inherit the security posture defined in the WIS_PT scene of the 1000-Year Future Map, enhanced with SROC-specific protections.

9.1 Zero-Trust Alignment

Every SROC operates under zero-trust principles. No actor is assumed trustworthy; trust is continuously earned through EarnedPath scoring and validated through ARI/RHC loops. Contract state transitions require multi-party cryptographic signatures on Gracechain.

9.2 5GW Awareness

SROCs are designed with fifth-generation warfare awareness. Attacks on land governance may come through information manipulation, ecological sabotage, economic subversion, or institutional capture. The SROC defense stack includes: NBERS anomaly detection (ecological attacks), Gracechain integrity verification (economic attacks), SOMT cross-jurisdictional correlation (institutional attacks), and RHC harmonic analysis (information-domain attacks that degrade resonance without triggering conventional alerts).

9.3 Temporal Security

The greatest security threat to a 1000-year governance system is temporal—the gradual erosion of constraints through precedent, exception, and institutional drift. SROCs counter this through: permanent-layer immutability on Gracechain, explicit decay timers on all temporary provisions, automatic re-validation cycles, and the SOMT's 1000-year viability score that flags slow degradation before it reaches critical thresholds.

10. Implementation Roadmap

Phase	Timeline	Deliverable	Dependency
Seed	Q2–Q3 2026	SROC specification v1.0; NBERS pilot scoring on 3–5 test tiles	NBERS Prospectus; first Pilot Quests (H1)
Root	Q4 2026 – Q2 2027	Gracechain testnet; SROC-P prototype contracts; UBIMIA distribution simulation	Meritcoin-Gracechain repo; PlayNAC-KERNEL alpha
Branch	2027–2028	SROC-U commons contracts for first watershed partner; SOMT tier integration (T1–T4)	Government Partnership Brief engagement; GSSG v3.0
Canopy	2028–2030	Full SROC-P/U/V deployment; SOMT planetary compilation (T5–T6); 1000-year viability scoring live	Multi-government adoption; SOMT Seed Capsule metadata
Forest	2030+	Interstellar SROC templates (T7–T8); generational transfer protocols; civilizational steady-state	GSSG interstellar architecture; SOMT full deployment

11. Formal Specification Summary

The following canonical entries define the SROC system for reference purposes.

SROC: Smart-Resonant Offset Contract. A cybernetically governed agreement validated through ARI/RHC resonance loops and recorded on Gracechain with cryptographic finality.

SROC-P: Personal Stewardship Contract. Governs household-tile relationships under BORDER_MM micro-rules.

SROC-U: Public Commons Contract. Governs non-alienable shared assets under WIS_PT permanent-layer constraints.

SROC-V: Private Venture Contract. Governs productive enterprise use under PROP_LL conditional stewardship.

RCS: Resonance Clearance Score. The ARI/RHC-derived score that determines whether a proposed SROC passes viability validation.

Offset Mode: The state entered when any SROC parameter deviates below specified thresholds. Triggers NPR remediation, not punishment.

Compilation Hierarchy: Public permanent → Private conditional → Personal micro. Inherited from How (ERES-HOW-ILO-2026-001).

Cybernetic Anchor: C = R × P / M. The SROC's method (M) must be clean enough that Resources × Purpose yield maximum Cybernetic output.

12. Conclusion

Smart-Resonant Offset Contracts represent the contractual realization of the ERES New Age Cybernetics vision: a world where every acre of land, every drop of water, every corridor of migration, and every act of enterprise is governed by agreements that resonate with the long-horizon viability of human civilization.

SROCs are not utopian abstractions. They are engineered mechanisms—computationally executable on Gracechain, economically self-financing through Meritcoin and UBIMIA, ecologically accountable through NBERS, and game-theoretically stable across infinite horizons. They do not require universal virtue; they require only rational self-interest operating within a properly designed incentive structure.

The 1000-Year Future Map provides the constraint. The How triad provides the ownership grammar. The SOMT provides the planetary loom. And the SROC provides the thread—the binding mechanism that weaves stewardship, dignity, and production into a fabric strong enough to carry civilization across the centuries.

The method is worthy of the purpose.

Build for generations to come.

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